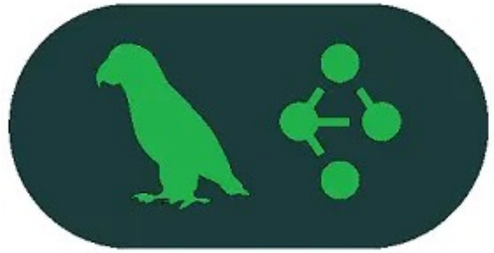
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From Foundations to Applications
Created by : Shreyas Shukla
in association with
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LangGraph

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What exactly is LangGraph ?



LangGraph

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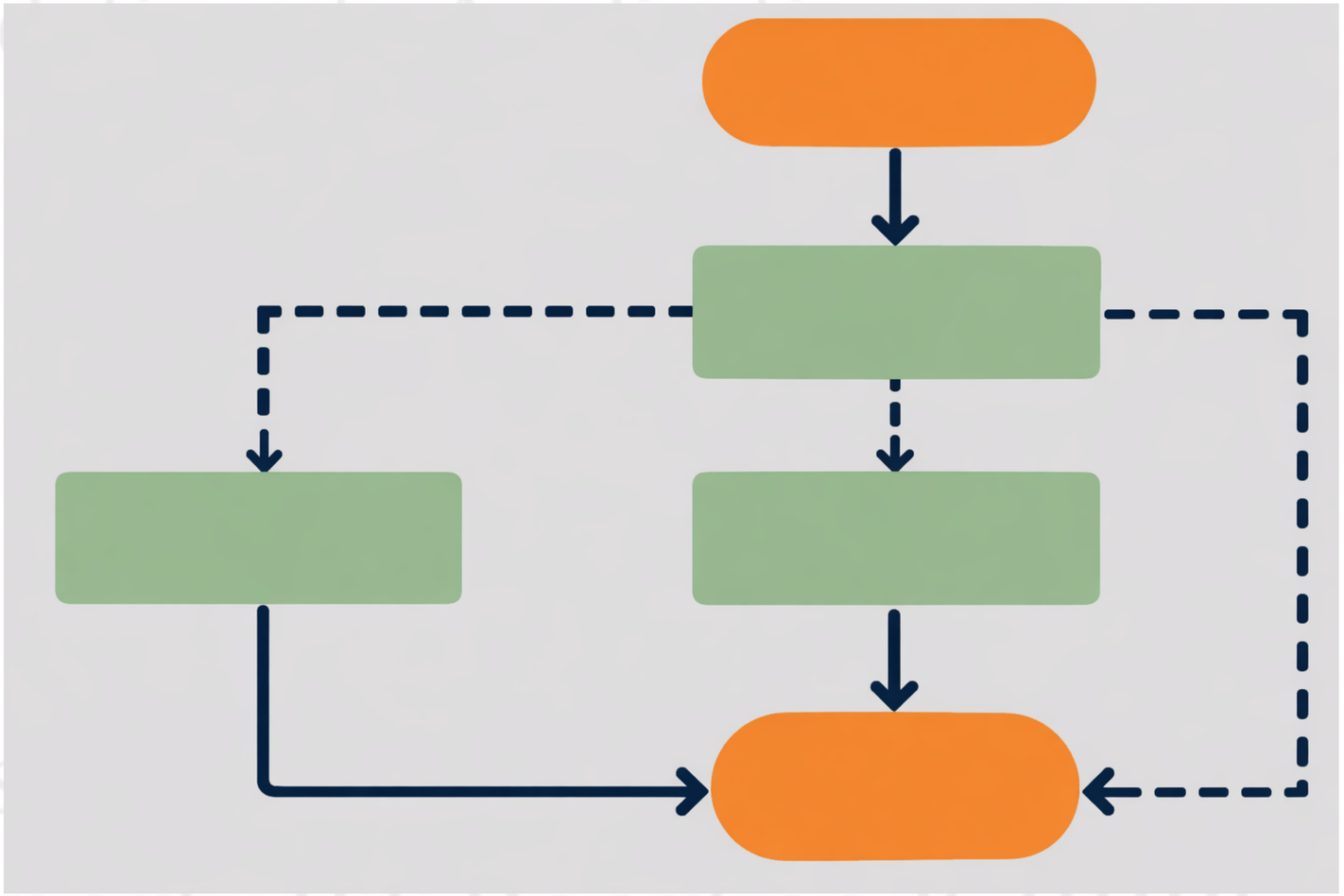
What exactly is LangGraph ?

“ Framework for building stateful, multi-step applications powered by LLMs “



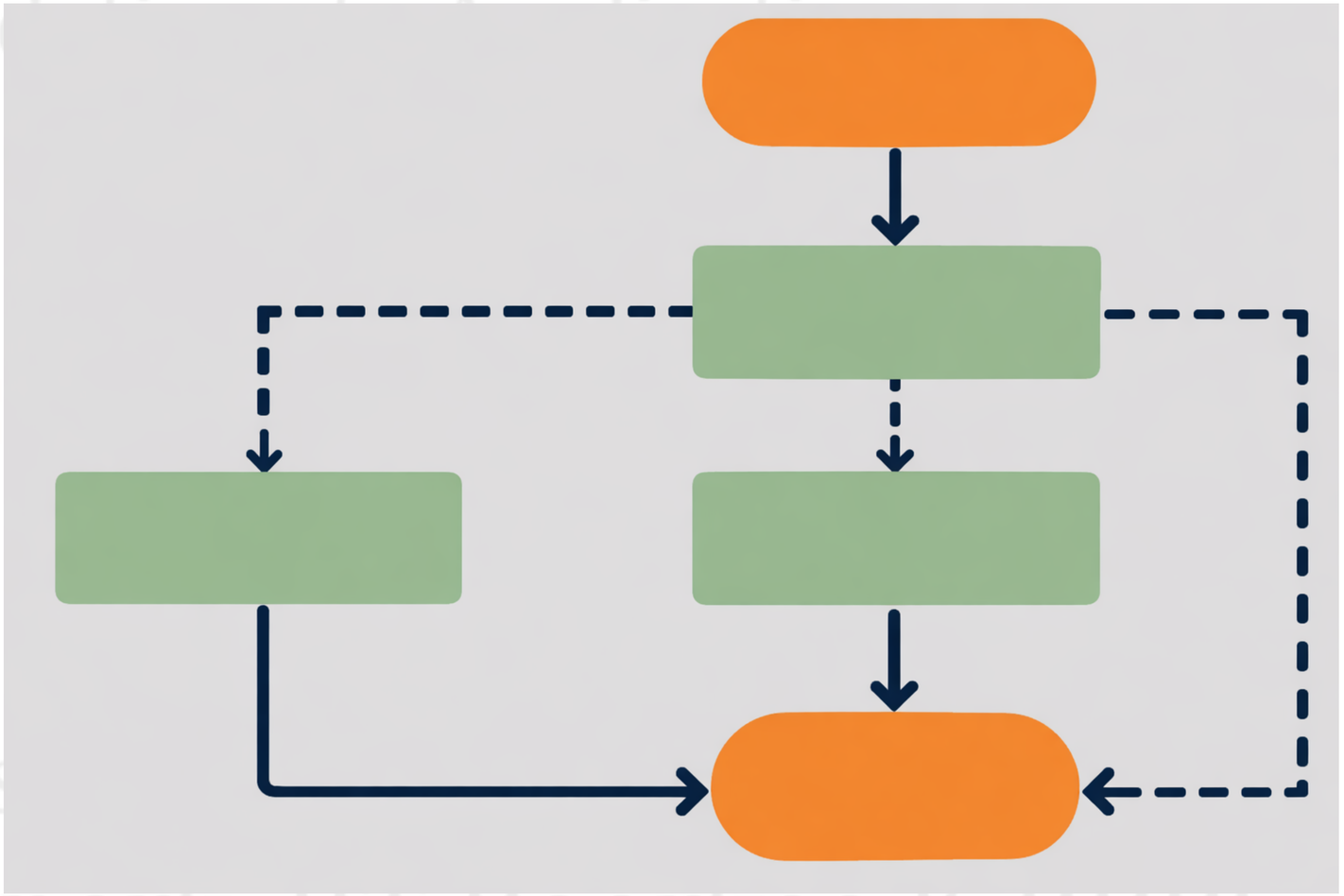
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- Decisions
- Memory
- Loops



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- Thinks
- Remembers
- Acts



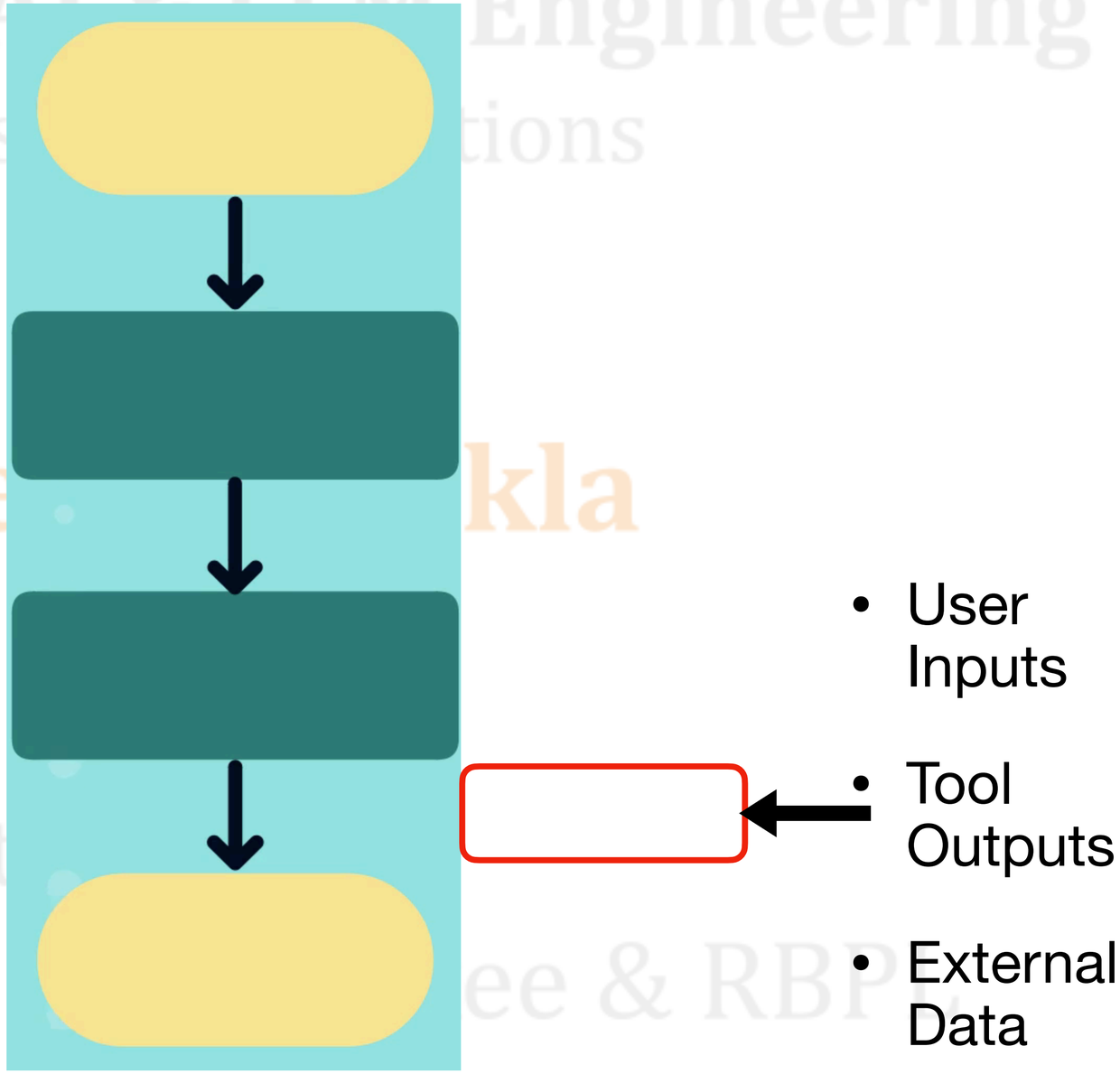
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Language Models are Stateless !!

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State

“ Information that travels from one node to the next, updating along the way”

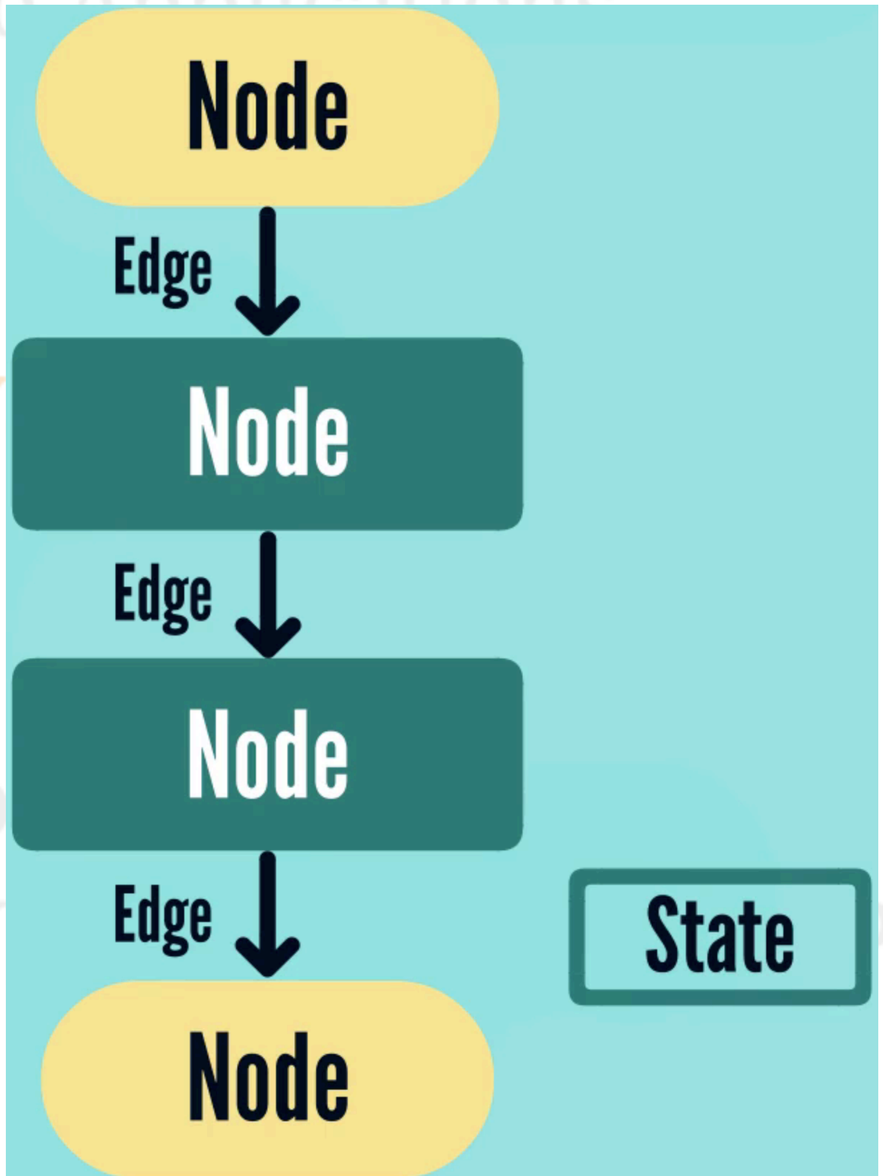


What is a Graph ?

1. States : Hold the data

2. Nodes : Perform operations or decisions

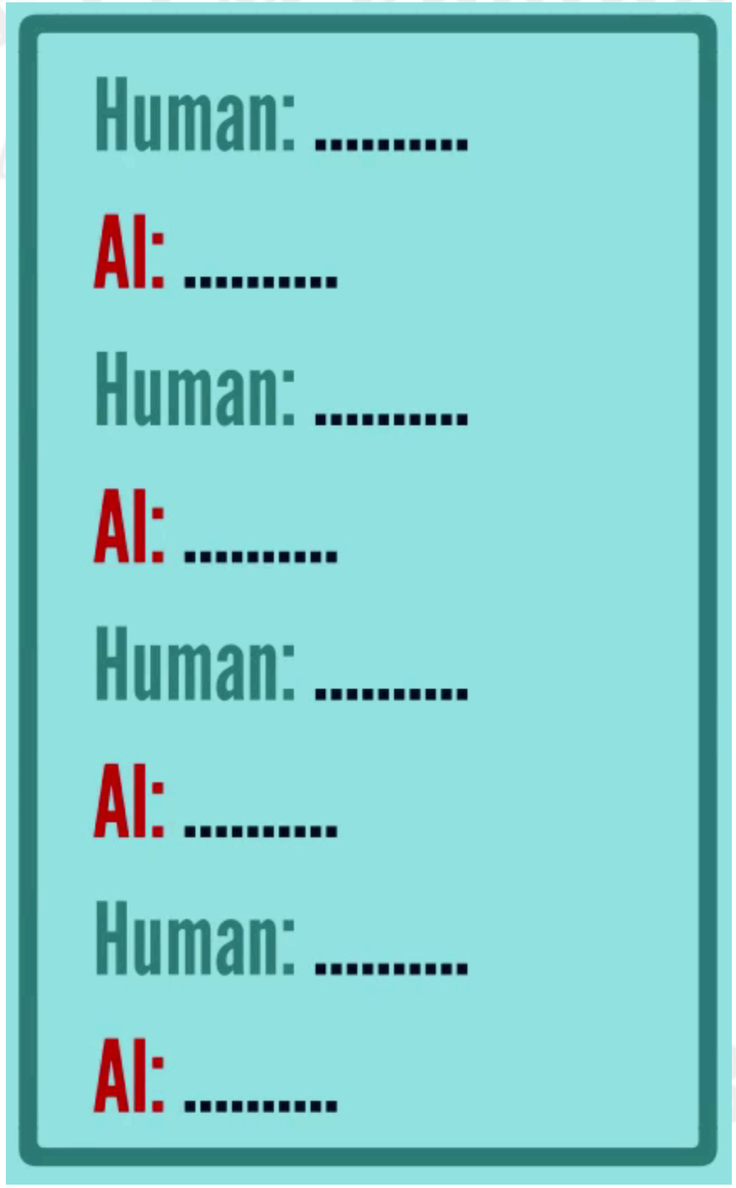
3. Edges : Connect the nodes



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Conversation Management

Chatbots should be
Smart and Efficient not
Forgetful and Bloated



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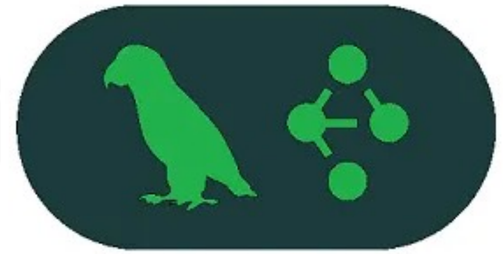
Memory Management

Short - Term Memory	Long - Term Memory
Persists context between graph runs	Stores data externally

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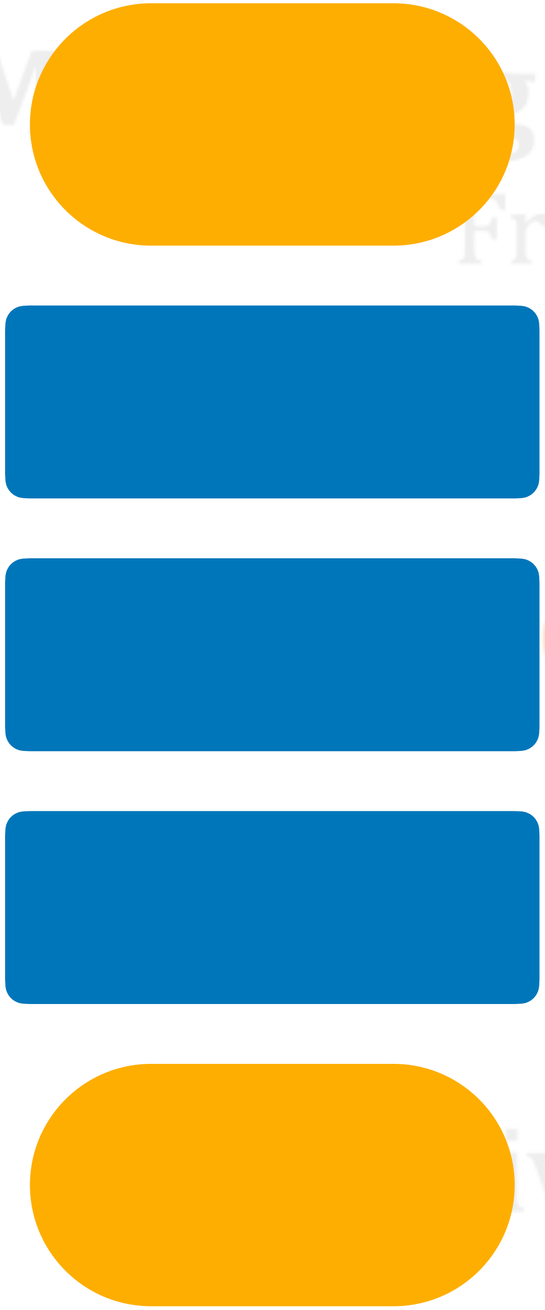
GRAPH



- **High Level Understanding**
- **Key Components**
- **Operations**

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GRAPH

- Defines an application’s flow
- Structured map of how information moves and how tasks are executed

Node

Node

Node

Node

Node

GRAPH
(Nodes)

User - defined Python functions that perform a specific task

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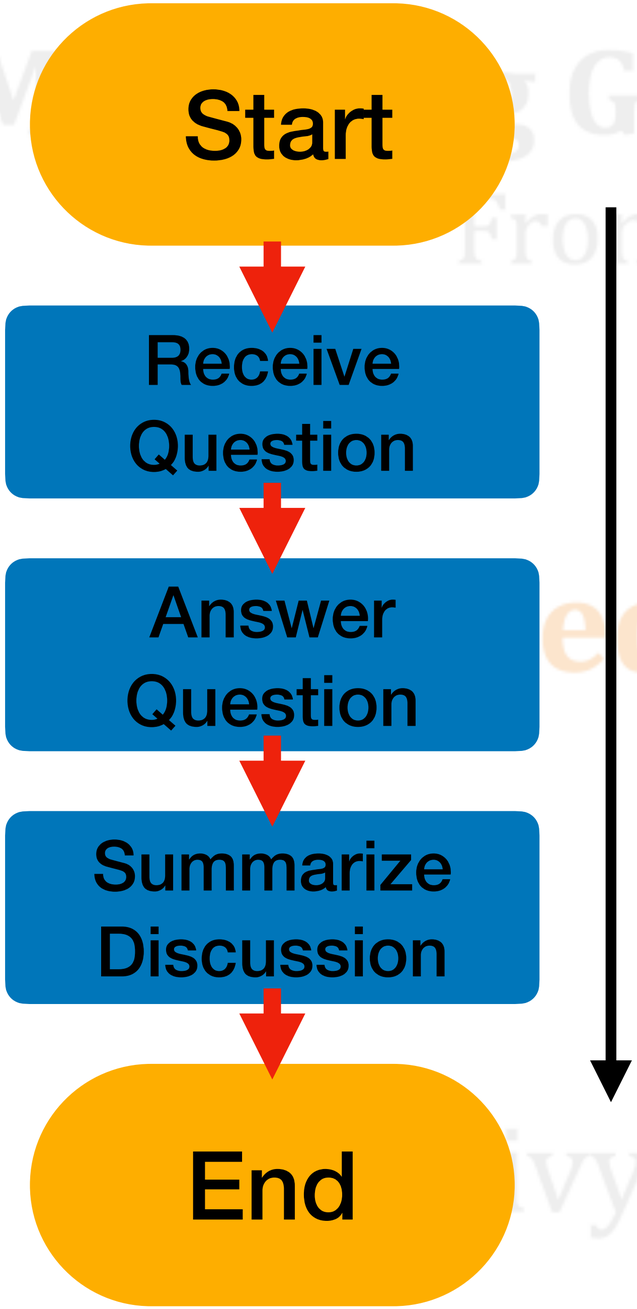
Start

Receive Question

Answer Question

Summarize Discussion

End

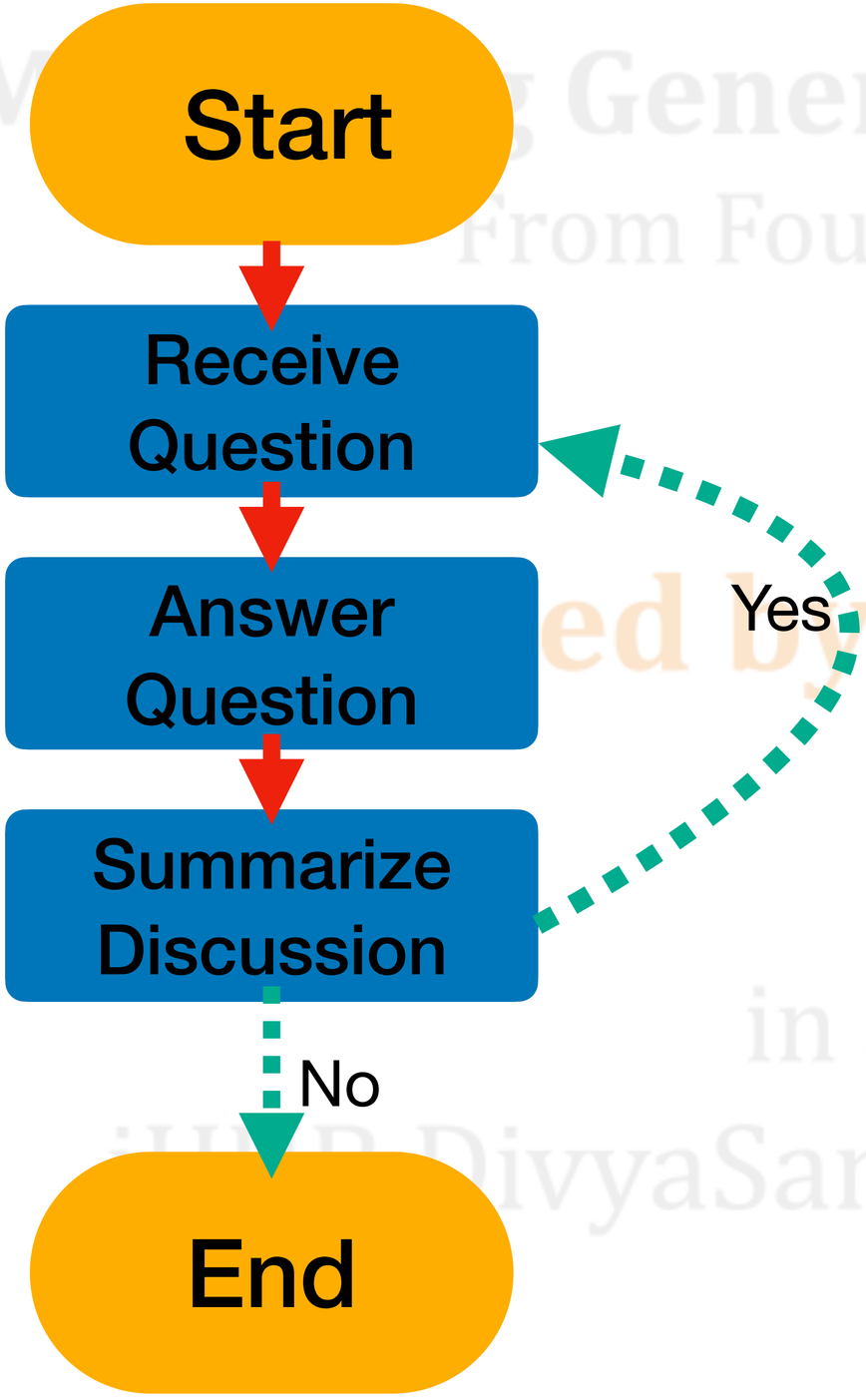


GRAPH (Edges)

Define how data flows from one node to the next

Python functions that decide which node to visit next

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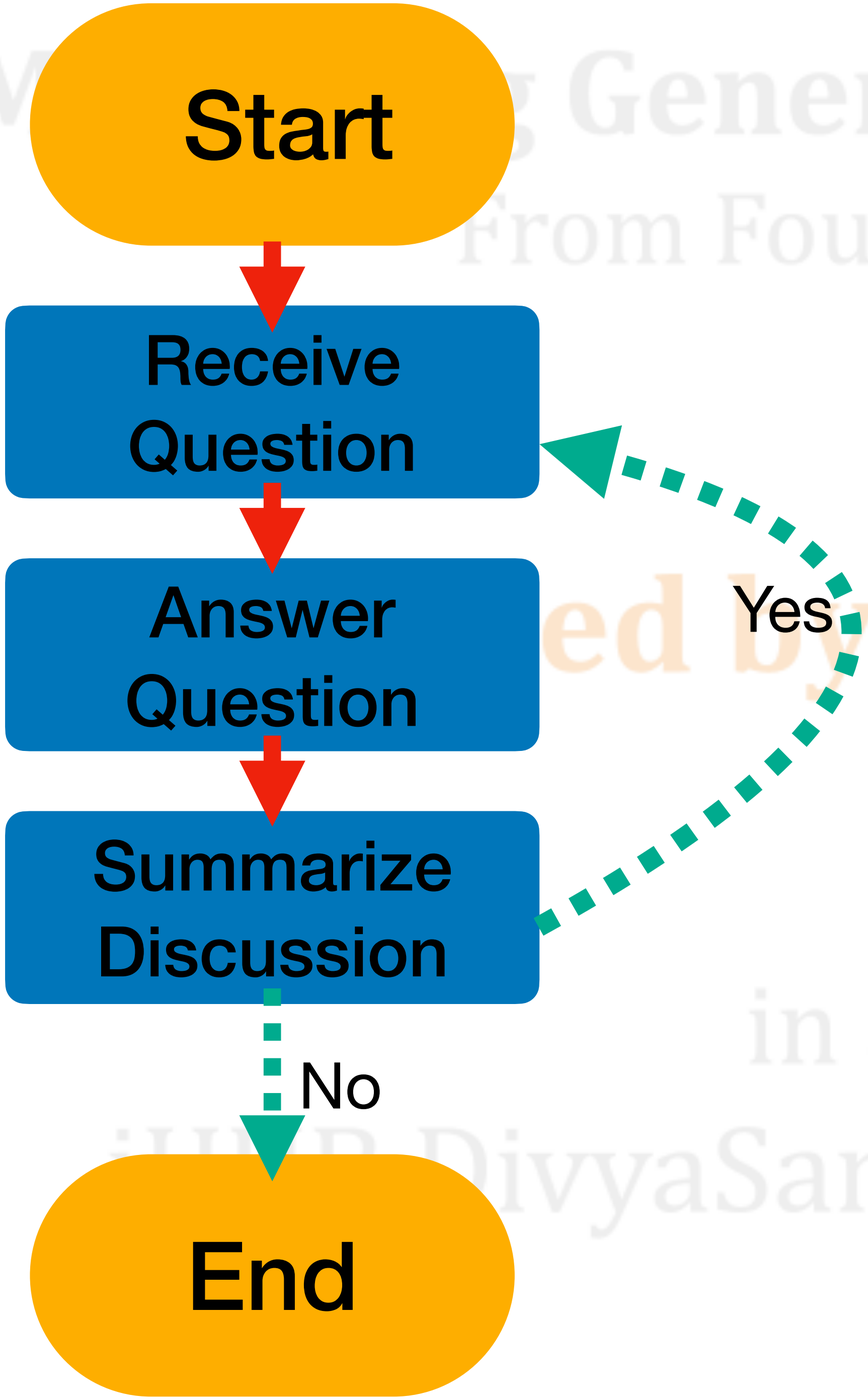


GRAPH
(Conditional Edges)

Route the flow differently based on a state value

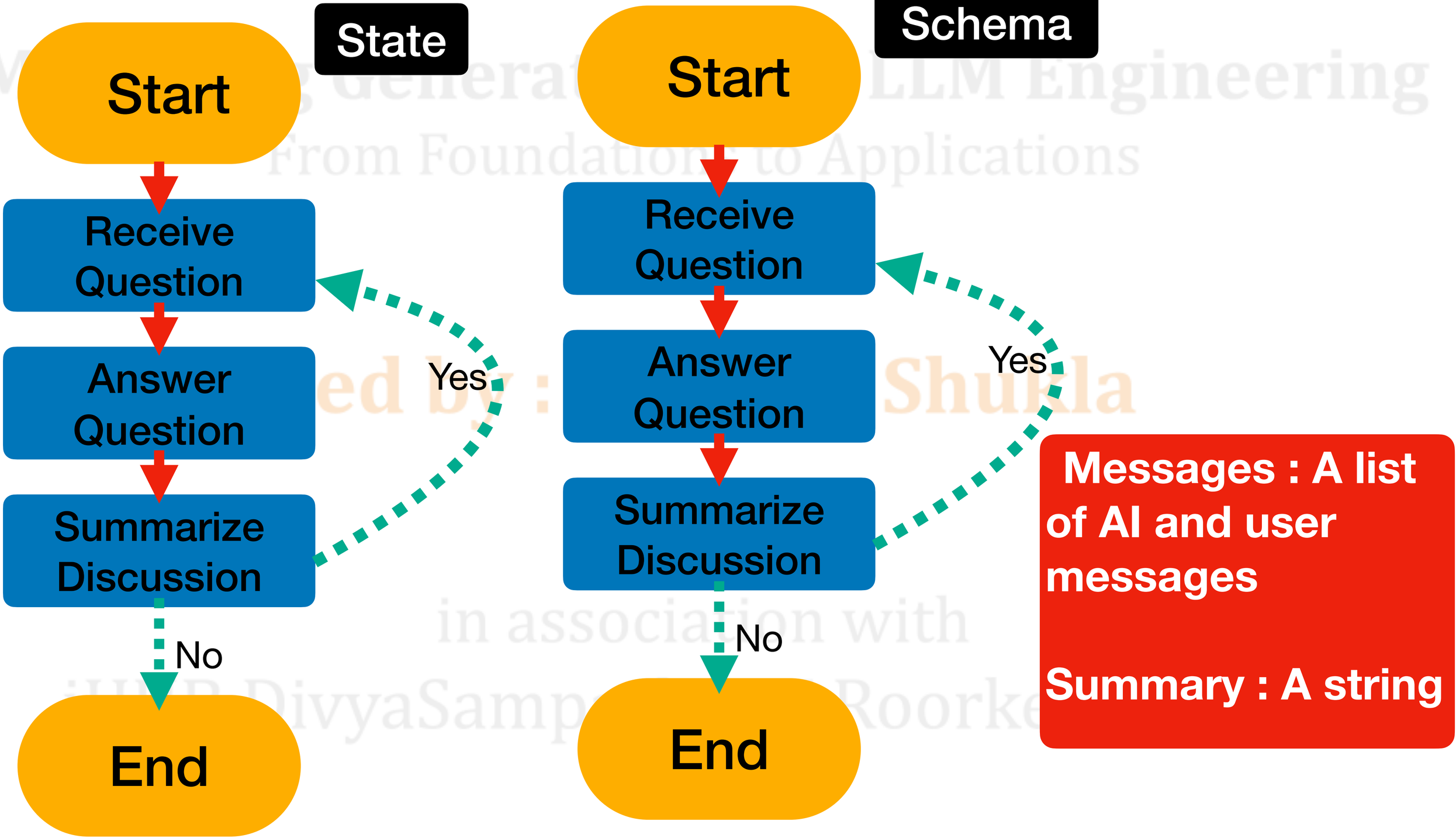
Allows for dynamic, adaptable applications

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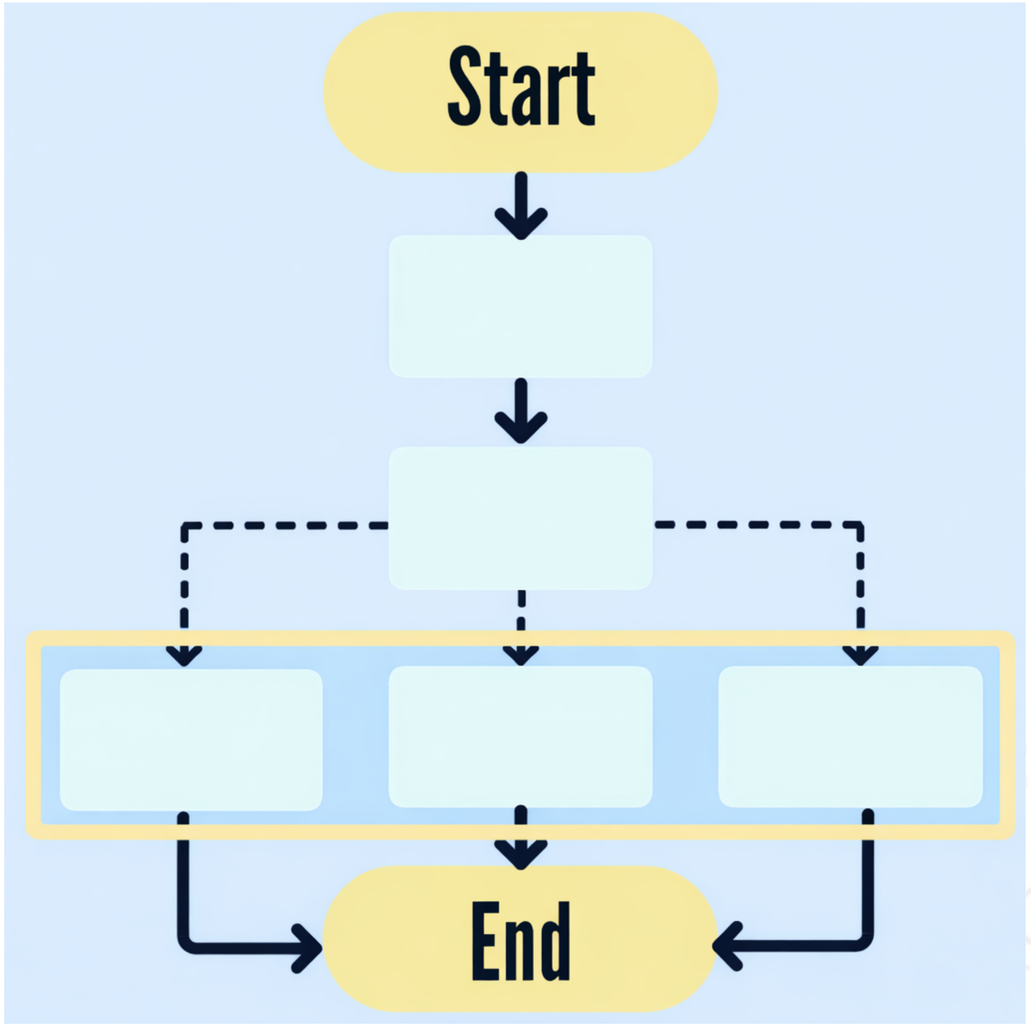
GRAPH
(State)

- **Represents shared data**
- **Updated as the graph progresses**
- **Defined by a schema - a Python dictionary (TypedDict or a Pedantic model)**


State
Start
Receive Question
Answer Question
Summarize Discussion
Yes
No
End
Schema
Start
Receive Question
Answer Question
Summarize Discussion
Yes
No
End
Messages : A list of AI and user messages
Summary : A string

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SuperStep



- Represents a single “tick” of the graph.
- Multiple nodes can be evaluated and executed in parallel.

A node only becomes active if it receives an input state, which is determined by edges leading into it